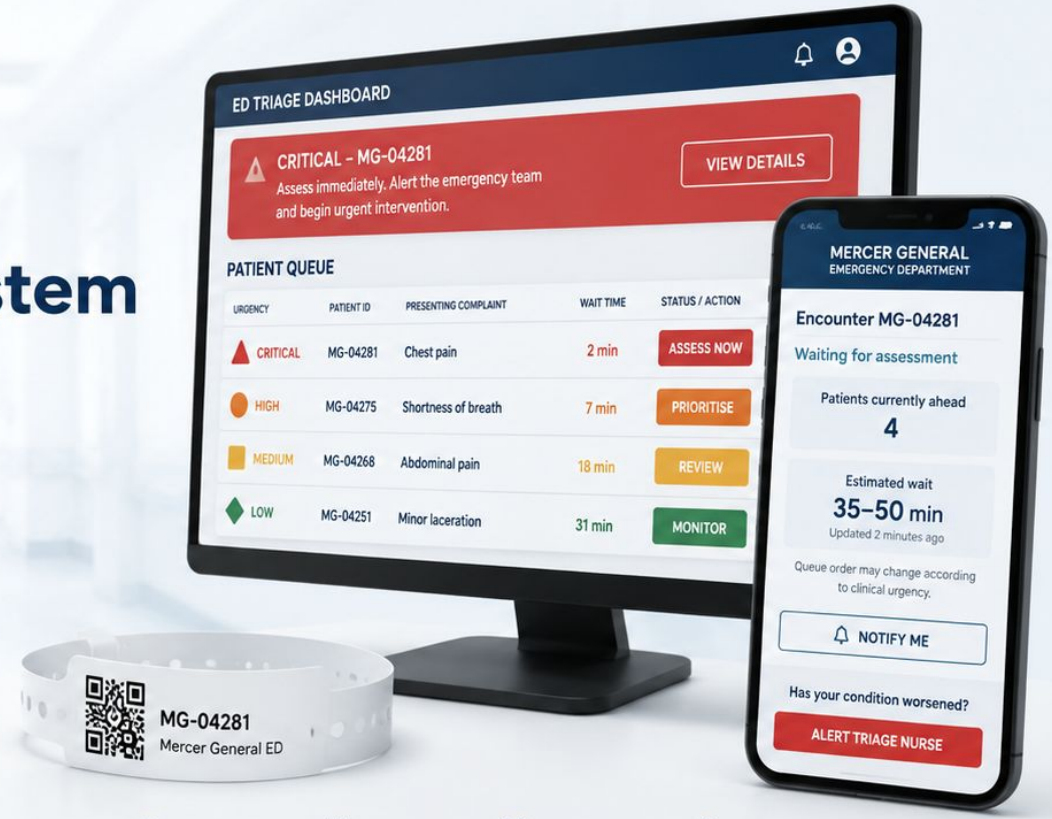
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Mercer ED
AI-Assisted Triage System

Final Integrated Human–Computer Interface Prototype

Combining my individual HCI proposal with the wider system developed by my team

Ansarah Mohammed
CariSurg MedTech Pathways
Week 10 – Deployment, System Integration & User Testing



Patient-Centred Care | AI-Supported Decisions | Clinician Authority | Safety & Accountability

Sensors inform. AI assists. Clinicians decide.

SLIDE 2 INTEGRATED HCI CONCEPT

Combining Individual and Team Design Concepts

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This final HCI prototype combines my individually proposed patient- and nurse-facing interface with the wider AI-assisted triage system developed by my team.

My Individual HCI Concept

Patient + Nurse Interaction

- Patient-facing waiting and queue support
- QR-accessible waiting portal
- Deterioration / reassessment request
- Nurse-facing patient queue
- Four urgency-tier alert system
- Short, actionable alert messages
- View Details interaction
- Accessibility and alarm-fatigue considerations

Team-Proposed System

Clinical + AI Integration

- QR-linked patient encounter
- Intelligent sensing-chair concept
- Structured clinical data collection
- AI-supported ESI recommendation
- Friction Protocol to reduce automation bias
- Explainable AI outputs
- Clinician confirm/override workflow
- Safety Hold for incomplete data
- Audit trail

Final Combined HCI

Patient Waiting HCI
QR portal for queue status, updates and reassessment requests

→

Clinical Data Collection
Triage intake, vitals, complaint and risk flags

→

Independent Nurse Assessment
Clinician records initial ESI before viewing AI support

→

AI Decision Support
Explainable urgency recommendation with key contributing factors and confidence

→

Confirm / Override
Clinician confirms or overrides with justification

→

Updated Queue & Clinical Record
Queue order updates, audit trail recorded, shared across the department

The combined design retains the patient- and nurse-centred HCI developed individually while incorporating the team's wider clinical integration and AI-safety mechanisms.

Sensors inform. AI assists. Clinicians decide.

SLIDE 3 PATIENT HCI – WAITING PORTAL

Patient HCI – Waiting Without Losing Contact

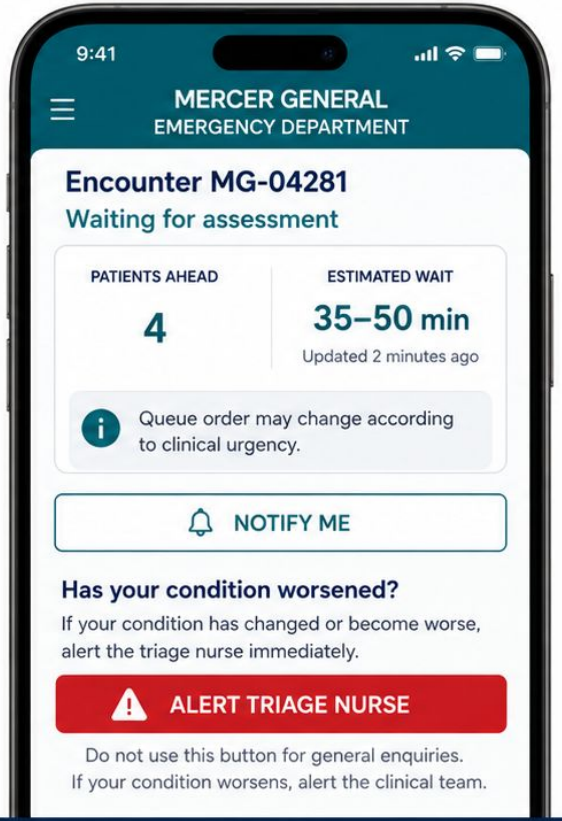
Patients can scan their encounter QR code to monitor their waiting status without repeatedly returning to the triage desk.

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Why this matters

- Supports patients waiting outside the main waiting area
- Reduces repeated queue-status enquiries at the front desk
- Provides notifications when it is time to return
- Offers a safe way to report deterioration while waiting
- Maintains connection with the ED and clinical team

The portal does not display or allow patients to change their clinical urgency tier.



How it works

Scan your QR code
Use your phone camera to scan the QR code on your wristband.

View your status
See your place in the queue and estimated waiting time.

Get notified
Receive an alert when it is time to return to the waiting area.

Report if worse
Request reassessment if your condition deteriorates while waiting.

Important
If you feel severe chest pain, difficulty breathing, fainting, or any life-threatening emergency, go to the ED immediately or call emergency services.

SLIDE 4 PATIENT HCI – REASSESSMENT REQUEST

Reporting Deterioration While Waiting

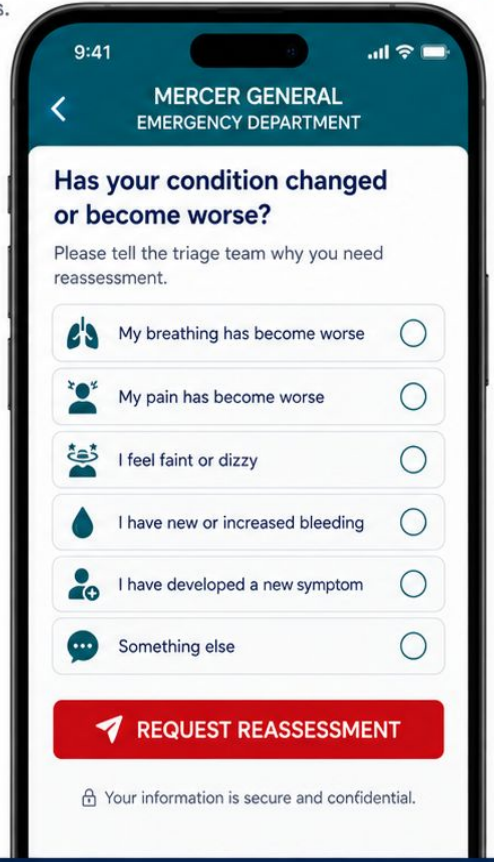
Patients can request a reassessment if their condition changes.
The clinical team determines urgency.

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How it works

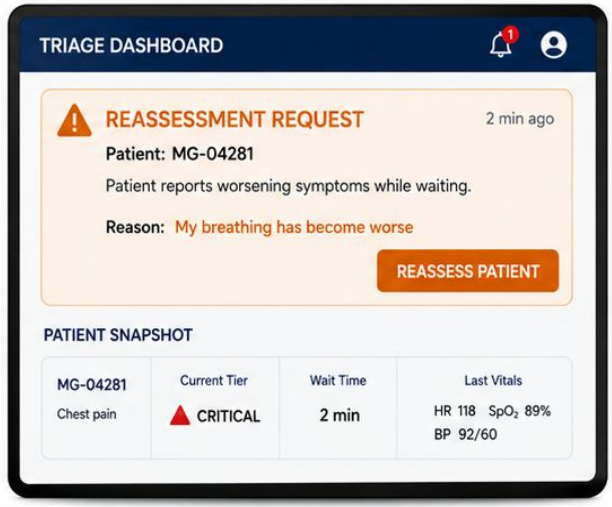
1. **Patient selects a reason**
   They choose the best option describing how they feel.
2. **Request is sent securely**
   The message is sent to the triage team in real time.
3. **Nurse is notified**
   The triage dashboard shows a reassessment request alert.
4. **Patient is reassessed**
   A clinician reviews the patient and determines any change in urgency.

This request does not change your urgency automatically.
A clinician will reassess you.



Request sent to triage team

What the triage nurse sees



All alerts are reviewed by a clinician.
The clinical team determines any change in urgency.

Nurse HCI – Acuity-Ordered Patient Queue

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The triage dashboard helps nurses prioritise patients quickly based on clinical urgency.

Key features

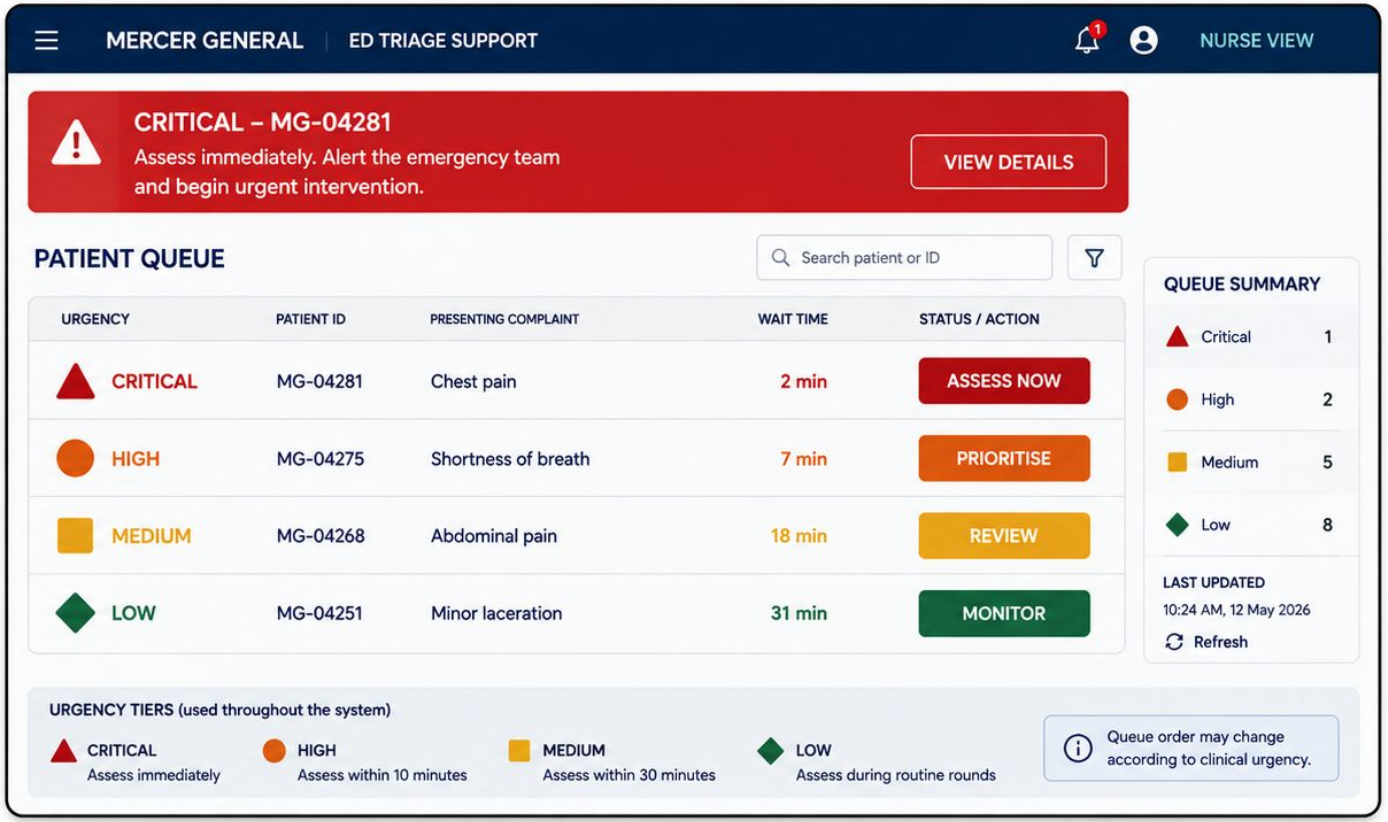
Acuity-ordered queue
Patients are sorted by clinical urgency, not arrival time.

Alert banner
The most urgent patient is highlighted with a clear, actionable message.

At-a-glance information
See complaint, wait time and next action for each patient.

Status & action
Clear actions help the nurse decide what to do next.

Critical and High alerts receive greater visual emphasis.
Medium and Low remain static to reduce alarm fatigue.



Clinician First – Assessment Before AI Support

To reduce automation bias, nurses record their independent clinical assessment before viewing AI recommendations.

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Why this matters

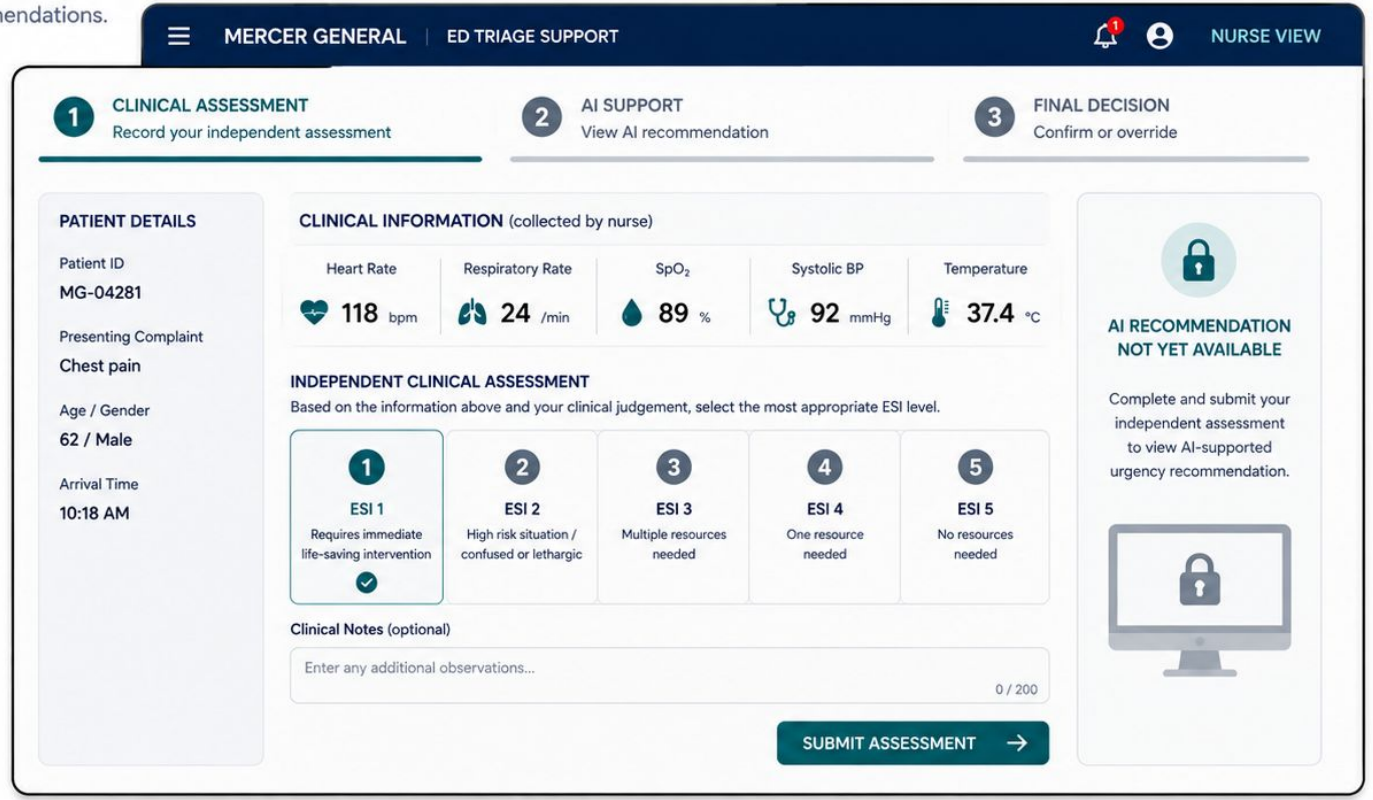
Helps prevent automation bias
Ensures clinical judgement is independent of AI suggestions.

Keeps clinicians in control
The nurse's assessment drives the first decision.

Improves decision quality
Encourages careful evaluation of the patient first.

Supports accountability
Initial assessment is recorded in the audit trail.

AI recommendations are hidden until the initial assessment is submitted.
This is our Friction Protocol.



Explainable AI Support – Not Just a Score

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After recording their independent assessment, nurses can view AI-supported urgency recommendations and the key factors that influenced the prediction.

Key features

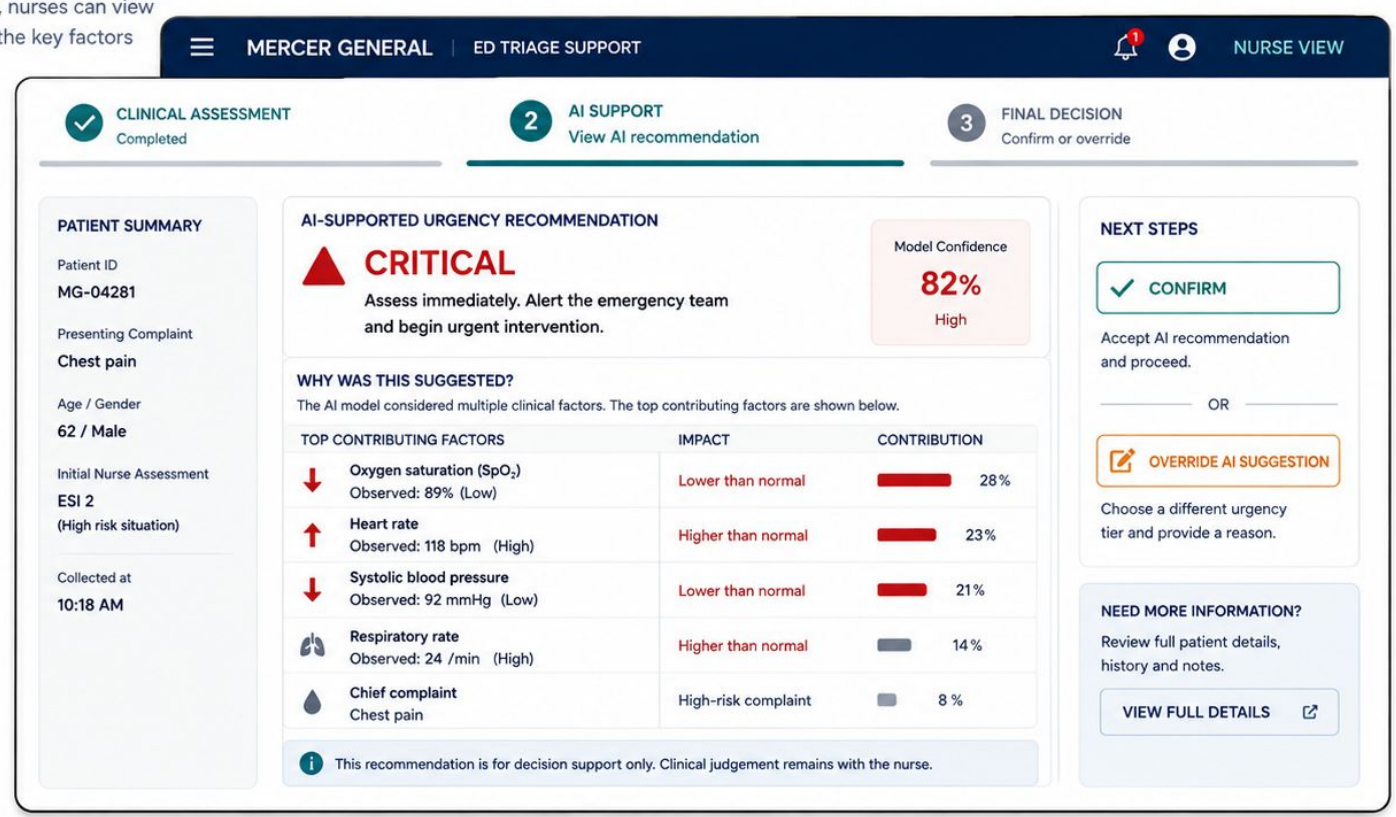
Explainable AI
Shows the top clinical factors influencing the prediction.

Confidence level
Displays model confidence to support appropriate trust.

Balanced decision-making
Supports informed decisions, not blind reliance.

Clinician authority
Nurses can always confirm or override the suggestion.

AI supports, clinicians decide.
The final decision is always made by the healthcare professional.



SLIDE 8 NURSE HCI – OVERRIDE & SAFETY HOLD

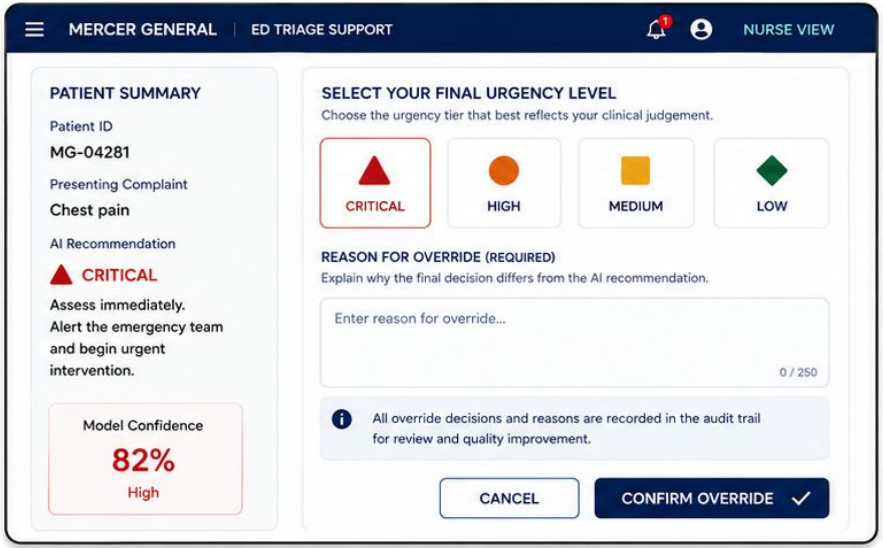
Clinician Authority & Safety First

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Clincal judgement remains with the healthcare professional.
Overrides are recorded with reasons. The system holds AI assessment when required information is missing.

A OVERRIDE AI SUGGESTION

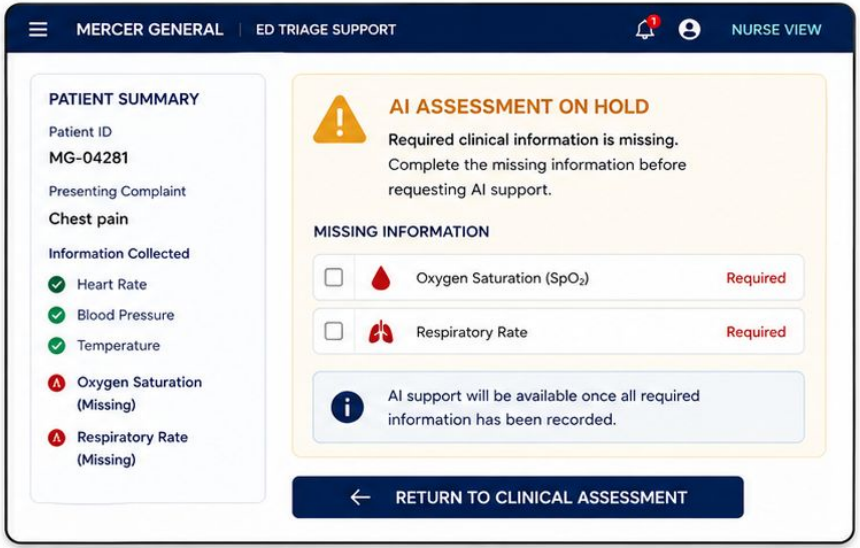
Clinicians can override the AI recommendation and provide a reason.



Your clinical judgement is the final decision.
The system supports you; it does not replace you.

B SAFETY HOLD – INCOMPLETE INFORMATION

When essential data are missing, the system abstains from AI prediction.



We do not guess.
When data are incomplete, the system holds AI assessment to maintain patient safety and decision reliability.

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Integrated System – From Patient to Clinical Record

All components work together to create a safe, efficient and transparent AI-assisted triage workflow.

System Components

Patient Interfaces
Waiting portal, reassessment request and queue display.

Clinical Data Collection
Intelligent chair and structured data capture.

AI Decision Support
Model processes data and provides ESI recommendation with explanations.

Nurse HCI
Acuity-ordered dashboard, friction protocol and override capabilities.

Clinical Record & Audit
Decisions, overrides and actions are logged for continuity and auditing.

End-to-End Workflow

1 Patient Arrival & Waiting



- Patient checks in and joins the queue.
- Can view position and general wait status.

2 Data Collection (Intelligent Chair)



- Vitals and basic clinical data are captured.
- Information is securely sent to the system.

3 AI Processing & ESI Recommendation



- AI model analyses structured and clinical inputs.
- Generates ESI tier, explanation and confidence level.

4 Nurse Assessment (Independent)



- Nurse records independent ESI assessment before viewing AI support.
- Follows friction protocol to reduce automation bias.

5 AI Support Review & Final Decision



- Nurse views AI recommendation, top factors and confidence.
- Confirms or overrides with reason.

6 Update Queue & Clinical Record



- Patient queue is updated in real time.
- Final decision and actions are saved to the EHR with full audit trail.

Continuous Feedback Loop

Data Security & Governance

- Data encrypted in transit and at rest
- Role-based access and permissions
- Audit trail for all decisions and actions
- Compliant with privacy and data regulations
- Explainability improves trust and accountability

Sensors inform. AI assists. Clinicians decide. | Technology supports the clinician, but the final decision always rests with the healthcare professional.

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Designed for Safety, Accessibility & Clinical Control

The Nurse HCI is built around core human-centred principles that support clinical judgement, patient safety and trust.

1. ACCESSIBLE COMMUNICATION

- Colour, shape and written labels are used together.
- Urgency is never communicated by colour alone.
- Clear, readable interface with high contrast and large text.
- Predictable layouts for quick recognition and understanding.

Information is easy to see, understand and act on for all users.

2. REDUCED COGNITIVE LOAD

- Short, actionable messages that drive quick decisions.
- Key information is prioritised and visible at a glance.
- Details are revealed on demand only when needed.
- Consistent structure reduces mental effort and errors.

Supports faster decisions, especially under pressure.

3. ALARM-FATIGUE REDUCTION

- Critical receives the strongest visual emphasis.
- High receives prominent but less intrusive attention.
- Medium and Low remain less intrusive.
- Alerts escalate predictably based on clinical urgency.

Helps clinicians focus on the patients who need it most.

4. HUMAN CLINICAL AUTHORITY

- Independent nurse assessment always comes first.
- AI recommendations are explained, not imposed.
- Clinicians can confirm or override with a recorded reason.
- Missing information triggers Safety Hold, not a guess.

Clinical judgment remains in control — technology supports, it does not replace.

NEXT STEPS

1. Finalise integrated model → 2. Conduct peer testing with ≥3 testers → 3. Analyse feedback (quantitative + qualitative) → 4. Refine HCI based on usability findings

Peer testing is pending completion of the final integrated prototype. Testing results will be documented using observed task performance and verbatim participant feedback to guide iterative design improvements.

Sensors inform. AI assists. Clinicians decide. | Safe. Explainable. Human-centred.